Performance assessment and Classification of Portuguese Water Utilities based on Non Revenue Water and Real loss indicators

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1 Introduction

Water is an essential resource to us, yet it faces significant global challenges. Therefore, effective water management is crucial to ensure sustainability and mantain high service quality. In Portugal, ERSAR monitors the effectiveness of water utilities through its Annual Report on Water and Waste Services. This report provides performance information for the country's water utilities, allowing them to identify underperforming indicators that could be improved.

One of the key performance indicators is Non-Revenue Water, which represents the volume of treated water that does not generate revenue. This volume includes unbilled authorized consumption, apparent losses and real losses. High levels of NRW indicate inefficiencies that result in economic losses and wasted energy, while elevated real losses additionally signal increased environmental impacts.

The aim of this project is to analyze ERSAR's data to assess NRW and real losses indicators across Portugal water utilities. Specifically, we aim to identify explanatory factors associated with these inefficiencies, investigate temporal trends in performance and classify utilities according to their performance levels.

2 Data

2.1 Preprocessing

The years selected for analysis were 2020 and 2023. In Excel, the variable `Ramo de Atividade` and the relevant variables and indicators were first selected. Subsequent data processing and computations were performed in `R`. For 2020, some indicators and variables were not available in the ERSAR dataset, namely:

- I_Ramais_afetados_por_falhas_no_abastecimento;
- V_Índice_de_segurança_e_resiliência;
- V_Comprimento_médio_de_condutas_com_mais_de_10_anos;
- V_Ramais_afetados_por_falhas_no_abastecimento;
- V_Volume_de_água_medido_nos_3_meses_com_maiores_volumes;
- V_Volume_de_água_medido_nos_3_meses_com_menores_volumes;

Additionally, the two indicators I_Resposta_a_reclamações,_sugestões_e_pedidos_de_informação_escritos and I_Resposta_a_reclamações,_sugestões_e_pedidos_de_informação_telefónicos appear to have been aggregated into a single variable, I_Resposta_a_reclamações_e_sugestões. Two classifications columns, RL_class and NRW_class, were created based on the respective indicators. Water utilities with missing values for I_Perdas_reais_de_água and I_Água_não_faturada were excluded from the analysis. In total, 22 utilities were removed from the 2023 dataset and 25 from the 2020 dataset.

2.2 Exploratory Data Analysis

The exploratory data analysis initially focused first on the 2023 dataset to identify potential explanatory variables and was subsequently complemented with the 2020 dataset to study temporal trends that could indicate improvements or declines in performance. A preliminary check for missing values revealed that, in 2023, three independent variables had over 60% missing values, while the remaining variables exhibited 30% or less. Consequently, the three columns V_Volume_de_água_medido_nos_3_meses_com_maiores_volumes, V_Volume_de_água_medido_nos_3_meses_com_menores_volumes and I_Resposta_a_reclamações,_sugestões_e_pedidos_de_informação_telefónicos were removed and imputation was performed on the remaining variables.

For the imputation, the median was used for all variables with missing values, with the procedure applied separately by Tipo_Sistema. Although the median is robust to extreme values and skewed distributions, imputation still changes the variable distribution. This effect was particularly noticeable in the interquartile range for variables with more than 10% missing values, but for indicators was less noticeable due to their derived nature. For variables and indicators related to connections, missing values were present in both types of systems. However, all bulk type utilities had no value for these

variables, whereas only some retail utilites were missing values. This is expected, as bulk utilities provide water to other entities rather than end consumers and therefore do not have connections. Thus, these missing values were left unchanged, as their absence reflects the structure of these utilities rather than a data deficiency.

The same approach was applied in the 2020 dataset. Overall, three columns with a high proportion of missing values were removed: I_Índice_de_valor_da_infraestrutura, V_Valor_atual_de_rede and V_Custo_de_substituição. The remaining values had 20% or less missing values so were subjected to median imputation. The first step in the analysis was to assess the distribution of both response variables, I_Água_não_faturada and I_Perdas_reais_de_água and to identify possible extreme values. It is important to note that only 10 of water utilities in the dataset are of the bulk type. Based on the boxplots presented in Figure 1, we can identify several outliers.

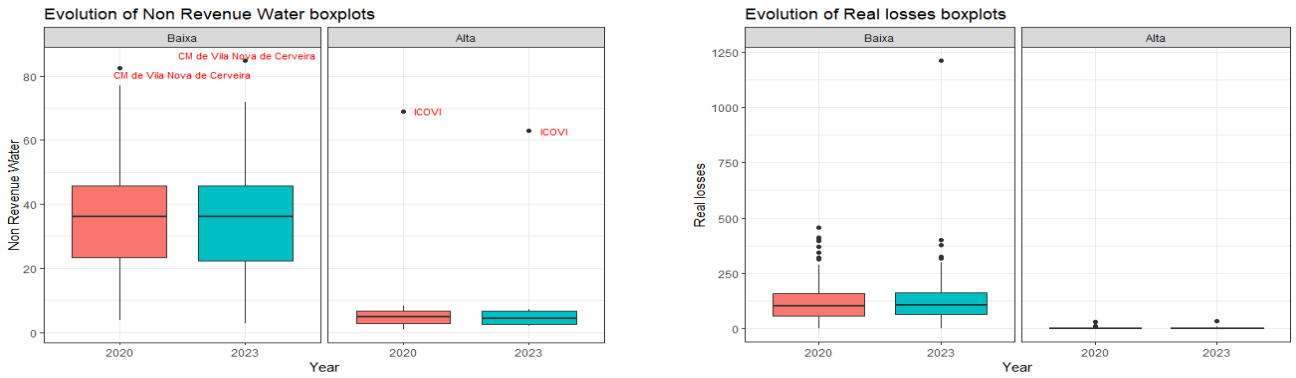


Figure 1: Distribution of non revenue water and real losses indicators throughout the years

For the bulk type utilities, ICOVI exhibited over 60% of NRW, whereas other entities had an average of approximately 10%. This high value is primarily due to unbilled authorized consumption. Among the retail utilities, CM de Vila Nova de Cerveira had over 80% of NRW, largely due to real losses. Interestingly, this utility wasn't flagged as an outlier for real losses. Indeed, in 2023, five other utilities were considered outliers for this variable alongside one bulk entitie EPAL. For instance, the entity INFRATRÓIA displayed a substancial discrepancy, with real losses exceeding double the values of others retail utilities. This high value appears to be related to the combination of pipes and connection failures and the lack of rehabilitation. After comparing these results with those from 2020 no major differences were observed between the two years for either indicator. However, for real losses, the previously identified outlier, INFRATRÓIA, stands out only in 2023, telling us that it suffered a significant deterioration between these years.

Even if no major differences were observed in the indicators, some water utilities may still exhibit notable variations in their values, resulting in shifts in performance. Analyzing the behavior of these utilities could provide valuable insight into factors contributing to improvements or deteriorations in NRW and real losses. The evolution of NRW was therefore studied and utilities with an absolute change greater than 15% were highlighted. A similar approach was applied to real losses, for which utilities showing an absolute change exceeding 75 liters per connection per day were identified. The corresponding results are presented in Figure 2.

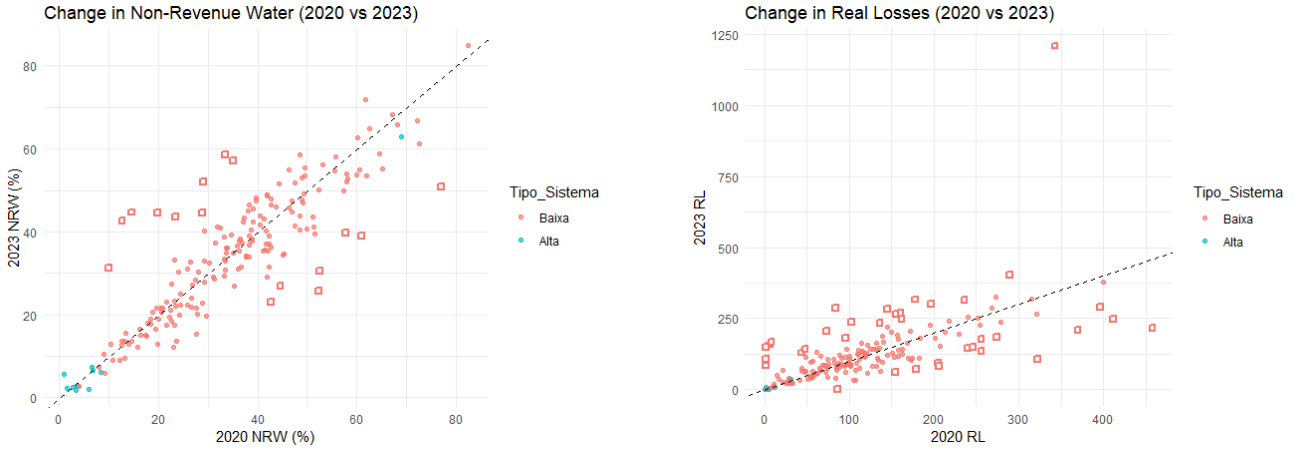
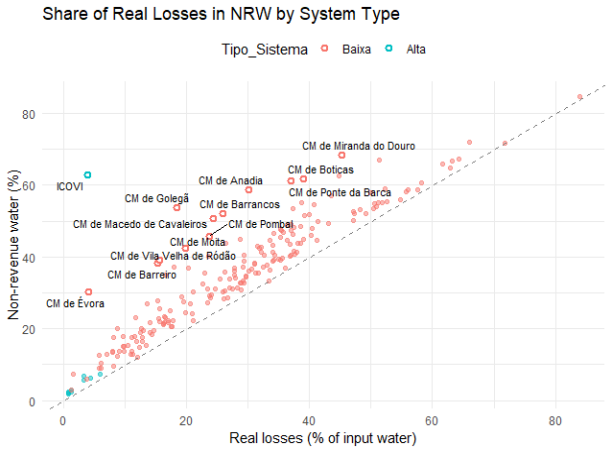
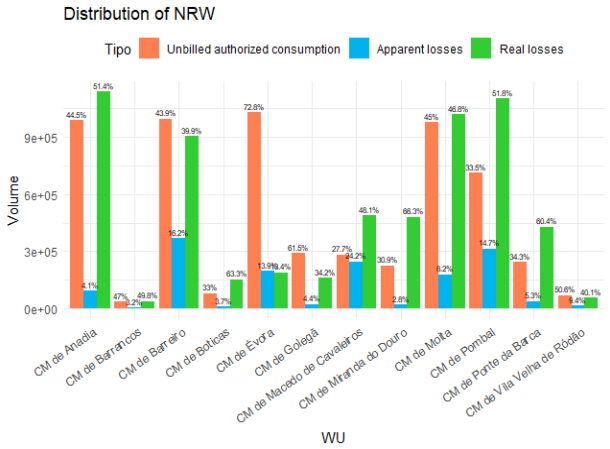


Figure 2: Evolution of non revenue water and real losses indicators

Another important aspect to analyze is the composition of the NRW across its three main components. We would like to determine whether utilities with a high percentage of NRW also exhibit high levels of real losses and, if not, to identify the sources contributing to the NRW. To this end, we identified utilities for which less than 80% of total NRW was attributed to real losses, as shown in Figure 3a. Overall, the majority of NRW is explained by real losses, as only 13 utilities were identified as exceptions and further examined. The percentage distribution of NRW across the three components for the retail entities in this group is presented in Figure 3b. Although all 13 were flagged, only five utilities exhibited a higher percentage of unbilled authorized consumption than real losses, indicating that in most cases, physical leakage remains the primer contributor to NRW.



(a) Relationship between NRW and real losses



(b) Composition of NRW across water utilities with less than 80% of NRW attributed to real losses

Retrieving the utilities previously identified as having large variations in NRW in Figure 2, we applied the same approach to examine whether the composition of NRW also changed over time. Overall, most water utilities that improved their NRW indicator exhibited either a decrease or stability in the proportion of real losses, whereas those that showed deterioration in NRW generally experienced an increase in this component, as shown in Figure 4. Among the entities with higher NRW values in 2023, the exception was CM de Barrancos, where the increase was mainly due to unbilled authorized consumption rather than real losses. In addition, for the utilities that improved their performance, the reduction in NRW appears to result from a combined effort to decrease both real losses and unbilled authorized consumption.

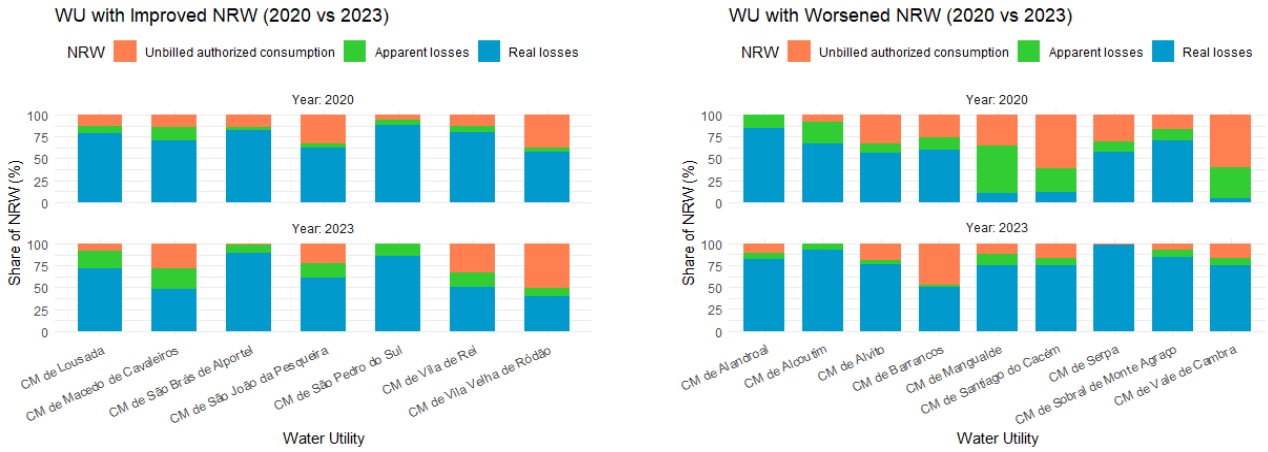


Figure 4: Evolution of composition of NRW for water utilities with an absolute change greater than 15%

Our analysis indicates that real losses are the primary contributor to NRW with unbilled authorized consumption accounting for only a small fraction in most utilities. However, the two indicators are not strongly correlated. This is because they measure different aspects. While NRW indicator reflects the proportion of system input non revenued, capturing overall performance, the real losses indicator reflects the utility's effectiveness in managing physical water losses. Thus, in terms of explanatory variables for NRW classifier we can conclude that variables related to network structure, rehabilitation or breaks and failures carry limited explanatory power as they substantialy overlap between classes. Similarly, absolute volumes of

losses, unbilled or billed consumption weren't as informative as we might think. Instead, ratio based variables that express each volume relative to system input showed a much clearer separation across classes, as shown in Figure 6.

After analyzing the dynamics of NRW, we now turn to the indicator for real losses. Although the volume of real losses represents a major component of the total NRW volume, these two indicators capture different dimensions of system performance. To begin, we examined the utilities previously identified in Figure 2 as having significant shifts in their real losses between 2020 and 2023. In total, 36 utilities were identified, of which only 8 also showed major variations in NRW over the same period. Among these 36 entities, 15 exhibited improvements in their real losses indicator. Next, among the retail entities in this group, we analyzed whether variables related network rehabilitation and operational failures had also undergone substantial changes. The results are presented in Figure 5.

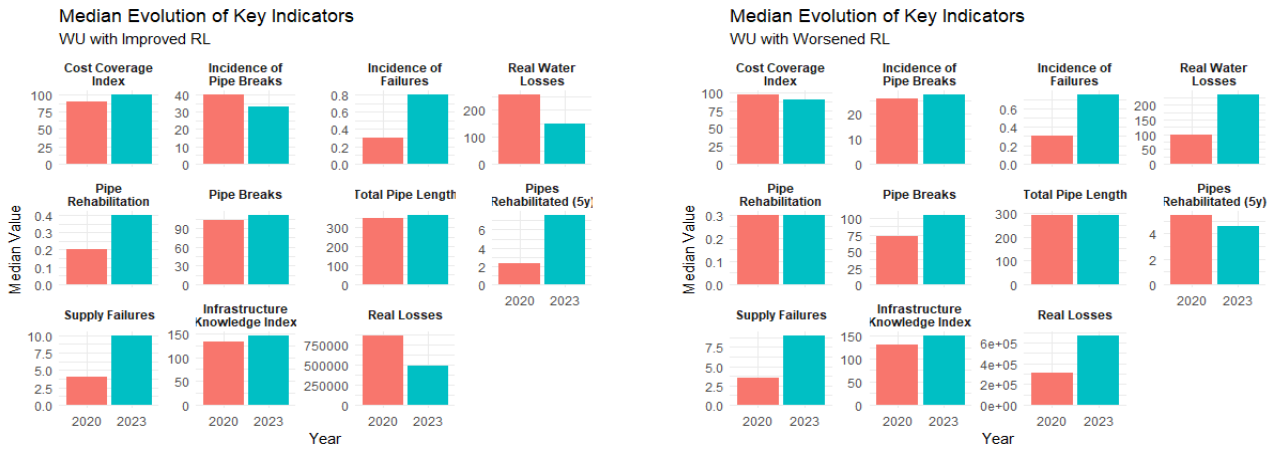


Figure 5: Differences in median value for water utilities with an absolute change of real losses excceding 75 l/(connection.day)

Overall, some variables remained relatively stable, while others experienced pronounced variation and across all utilities there was an increase in supply failures. In contrast, pipe rehabilitation activity showed divergent trends, as utilities that improved their real losses results exhibited an increase in this matter. These findings suggest that low frequencies of pipes breaks or supply failures alone are not sufficient to reduce real losses. Rather, it's the proactive effort in network rehabilitation that appear to play a bigger role in achieving lower water losses.

Even though rehabilitation variables appear to explain the major shifts observed in our dataset, their overall range remains quite low, indicating that most water utilities carry little to no rehabilitation. This limited variation reduces their explanatory power in distinguishing among the real losses classes. Moreover, rehabilitation efforts in 2023 may not have an immediate impact on real losses, since their impact is likely to appear gradually as the network improves over the next few years.

Take into account the results from Figure 7, the average length of pipes older than 10 years can be a potentially discriminant feature, particularly in separating the unsatisfactory class from the others. However, for most of the network structure and operational variables, such as pipe breaks and supply failures, a strong overlap across variables is evident. This suggests that a higher number of failures or leakages does not necessarily translate into greater water losses, possibly because utilities with efficient detection and repair protocols can mitigate losses effectively even under higher failure frequencies.

When examining the indexes, volumes and financial variables the separation between classes does not substantially improve. Both real losses volume and non revenue water percentage show some discriminative ability, particularly in distinguishing the Good class from the others. Also, the Infrastructure Knowledge Index also exhibits a similar trend, suggesting that utilities with better infrastructure monitoring tend to manage losses more effectively. Nevertheless, the financial variables, such as tariff revenues and total expenses, continue to show strong overlap between classes, implying that economic performance alone does not capture efficiency in water loss management.

We also explored potential interactions between variables. For example, whether a combined effect of higher rehabilitation rates and fewer failures could better discriminate between real losses classes. However, no clear patterns emerged. This may be due to the low range and limited variability of certain variables, such as rehabilitation and high within class variance observed across utilities.

Overall, rehabilitation appears to explain the major shifts in real losses observed between years, but its explanatory power diminishes when generalized across all utilities. Some variables still show meaningful relationships with real losses, particularly those reflecting the age and management of the network as well as volume-based indicators like real losses and non-revenue water. These results suggest that while targeted rehabilitation efforts may drive localized improvements, broader patterns of performance are more strongly influenced by long-term infrastructure conditions and operational efficiency.

3 K-Nearest Neighbors (KNN)

To address the classification tasks for *Non-Revenue Water (NRW)* and *Real Losses (RL)* among retail utilities, we implemented a K-Nearest Neighbors (KNN) classifier. For each classification problem, models were trained *with* and *without* the key variables defined in the project assignment (i.e. `v_agua_entrada_no_sistema` and `v_agua_nao_faturada` for NRW, and `v_comprimento_total_de_condutas`, `v_ramais_de_ligacao` and `v_perdas_reais` for RL).

3.1 Preprocessing

The dataset was filtered to include only retail entities and to remove columns that were not suitable as predictors. Specifically, identifiers (i.e. `empresa`, `tipo_sistema`, `ramo_atividade`, `identificação_da_entidade_gestora`) were removed from the predictor set to prevent data leakage and overfitting. These identifiers are unique to each utility and allow the KNN, which is a distance-based classifier, to memorize each entity-label pair instead of learning patterns and relationships between operational indicators and variables, and the outcome.

Additionally, identifiers, specially high-cardinality identifiers such as `empresa`, expand into hundreds of one-hot dummy columns, resulting in extremely sparse and high-dimensional vectors. Since KNN aggregates distances across all dimensions, with these sparse and high-dimensional vectors, distances are dominated by numerous level-mismatch penalties that are unrelated to the real drivers of NRW and RL, leading to nearest and farthest neighbors becoming similarly distant, as well as making neighbor selection unstable and largely determined by arbitrary codes rather than informative indicators. Therefore, removing pure identifiers prevents this distortion, reduces variance and computational cost, and improves generalization.

Lastly, other indicators and variables were removed manually. Non-numeric missing flags such as "NR" were treated as missing values (i.e. "NA"). Predictors that had over 50% of missing values were discarded, while predictors with missing values between 20% and 50% were judged case by case. Specifically, the predictors `i_ramais_afetados_por_falhas_no_abastecimento` and `v_ramais_afetados_por_falhas_no_abastecimento` presented missing values within this range. As these two variables were strongly associated ($r \approx 0.996$), they conveyed essentially the same information. Nevertheless, due to their relevance, only the latter was dropped to prevent redundancy while preserving representativeness. Similarly, `v_valor_atual_da_rede` and `v_custo_de_substituicao` also exhibited missing values between 20% and 50%. Both variables were considered conceptually relevant to the infrastructure valuation dimension, however, given their high correlation ($r \approx 0.924$), retaining both would have introduced multicollinearity. Consequently, `v_custo_de_substituicao` was discarded, as `v_valor_atual_da_rede` offered broader informational content by capturing the network's depreciated value, which implicitly reflects replacement costs. Additionally, the variable `i_indice_de_valor_da_infraestrutura`, despite having a similar proportion of missing values (20–50%), was preserved due to its analytical importance as a composite indicator summarizing infrastructure condition, deemed essential for model interpretation. Finally, the following predictors were dropped, as they had over 50% of missing values.

- `v_volume_de_agua_medido_nos_3_meses_com_maiores_volumes`;
- `v_volume_de_agua_medido_nos_3_meses_com_menores_volumes`;
- `i_resposta_a_reclamacoes_sugestoes_e_pedidos_de_informacao_telefonicos`.

All instances of "NA" were replaced with the median and mode in numerical and categorical predictors, respectively.

We used `recipe` to ensure reproducible preprocessing:

- Removal of near-zero-variance predictors:

  Predictors that are almost constant were dropped, because KNN is sensitive to noisy/irrelevant dimensions;

- Median imputation for numeric features and mode imputation for categorical features:

  Median and mode were chosen for their robustness, stability and the fact that they do not assume any distribution;

- Grouping of rare levels (`step_other`) and handling of novel levels in new data (`step_novel`):

  Levels that appear in less than 2% of rows were grouped into an "other" bucket to control the explosion of dummy columns and improve neighbor stability. A "novel" level placeholder was prepared for categories not seen during training to prevent errors when new data contains a category that never appeared in the training folds;

- One-hot encoding for categorical variables (`step_dummy`) and drop now-constant predictors (`step_zv`):

  Turns categorical features into 0/1 dummy columns since KNN needs numeric features, and removes any predictors that became constant after dummying;

- Class balancing via SMOTE to mitigate class imbalance in the outcomes:

  KNN is very sensitive to class imbalance and, without SMOTE, nearest neighbors skew to the majority class, hurting minority recall and balanced accuracy;

- Normalization (centering and scaling) of numeric predictors:

  Centers and scales numeric values to mean 0 and standard deviation 1, because distance-based methods must put all features on comparable scales to avoid large-scale variables dominating distances.

In line with the assignment, each classifier was run twice, with and without the key variables.

3.2 Model Specification and Tuning

In the execution of the KNN algorithm, different parameter configurations were tested. The `weight_func` parameter was varied as follows:

- `"rectangular"`: assigns equal weights to all neighbors.
- `"inv"`: assigns higher weights to closer neighbors, increasing sensitivity to local distances.

Similarly, two distance metrics were compared:

- **Euclidean distance:** `dist_power = 2` (more sensitive to outliers and amplifies large differences).
- **Manhattan distance:** `dist_power = 1` (more robust in high-dimensional or dummy-variable-rich datasets).

The cross-validation process selected the value of k that maximized Balanced Accuracy on the validation folds. The number of neighbours, $k \in \{3,\ldots,45\}$, was tuned through stratified v-fold cross-validation (default $v = 5$), with v automatically reduced if required by the smallest class size to ensure that all classes were represented in each fold. Balanced Accuracy was chosen as the primary selection metric, since it avoids bias towards the majority class and provides a fairer evaluation of the model's overall discriminative ability. The finalized workflow was then refitted on the full training data and evaluated on the independent test set.

3.3 Performance Metrics

Performance was reported using several complementary metrics:

- **Accuracy** – Measures the overall proportion of correct classifications. However, it may be misleading in imbalanced datasets;
- **Balanced Accuracy** – Mean of the recall across all classes. It ensures equal contribution from each class and robustness to imbalance.
- **Cohen's κ** – Assesses agreement between predictions and true labels, adjusted for chance agreement:

Table 1: Interpretation of Cohen's Kappa

Kappa Range	Interpretation
< 0	Worse than random
$0.00 - 0.20$	Slight agreement
$0.21 - 0.40$	Fair agreement
$0.41 - 0.60$	Moderate agreement
$0.61 - 0.80$	Substantial agreement
$0.81 - 1.00$	Almost perfect agreement

- **F-measure (macro)** – Harmonic mean of precision and recall, averaged equally across classes. A value close to 1 indicates balanced precision and recall for all classes, while low values reflect poor performance in at least one class.
- **Multiclass Log-Loss** – Evaluates the quality of predicted probabilities rather than discrete classifications. Low log-loss indicates confident and correct predictions (a random classifier would yield log-loss ≈ 1.10 for three classes), while high values indicate poor calibration or confident misclassifications.

3.4 Results and Comparative Analysis

For each of the two classification tasks (NRW and RL), multiple KNN variants were executed under different weighting, distance, and preprocessing configurations, namely with the inclusion of PCA, the removal of predictors with correlation larger than 90%, and both. Due to the lack of space, only the comparative summaries of the best models will be shown below. All the comparative summaries are displayed in 7.2.

Table 2: Best Models for NRW and RL

Target	Weight	Distance	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Inverse	Manhattan	Included	15	0.780	0.625	0.842	0.739	0.590
RL*	Inverse	Euclidean	Not Included	31	0.575	0.389	0.711	0.582	0.936

*without highly correlated predictors.

From Table 2, it is evident that for NRW, the combination of inverse weighting with Manhattan distance and inclusion of key variables produced the best results, achieving 0.842 balanced accuracy, 0.739 F-measure, and a relatively low log-loss of 0.590, indicating both strong class discrimination and well-calibrated probabilities. The optimal k of 15 allowed effective local sensitivity while maintaining generalization. In contrast, RL attained its highest performance with inverse weighting and Euclidean distance, without highly correlated predictors, reaching 0.711 balanced accuracy, 0.582 F-measure, and a higher log-loss of 0.936, reflecting more difficulty in probability calibration.

Moreover, the analysis of the comparative summaries in 7.2 provided insights not only into the impact of key variables but also into the role of distance weighting and preprocessing strategies.

For **NRW**, the inclusion of key variables improved performance under inverse weighting and even more prominently with Manhattan distance. This configuration achieved the best overall results, with high balanced accuracy and strong class discrimination. The *Medium* class was the most negatively affected when key variables were removed.

For **RL**, the inclusion of key variables had a mixed impact, with it contributing to an improvement in the inverse-weighted Euclidean and Manhattan variant, as well as in the variant with both PCA and the removal of highly correlated predictors, having a negative impact on the other variants. The *Medium* class remained difficult to separate in all settings, possibly due to gradual or overlapping boundaries in real operational performance.

Regarding preprocessing, PCA often led to performance deterioration. By projecting data into non-interpretable components, PCA reduced the model's discriminative capacity, especially when key variables were included, leading to sharp drops in κ and balanced accuracy and higher log-loss. This suggests that the original variable structure contains essential relationships for class differentiation.

Conversely, removing highly correlated variables (`step_corr`) yielded more balanced results, sometimes improving performance, particularly for RL. The combination of PCA and correlation filtering, however, proved overly aggressive, often eliminating relevant information and producing poorly calibrated probabilities.

Distance metric choice revealed a clear pattern: Manhattan distance consistently outperformed Euclidean distance for NRW, whereas RL achieved more stable results with Euclidean distance, reflecting differences in variable scaling and distribution.

Finally, the optimal number of neighbors (k) varied substantially across configurations, reflecting the interaction between feature space structure and distance weighting. In general, smaller k values yielded better performance when key variables were included, since these provided strong local discrimination, whereas larger k values were required when they were excluded, to counterbalance increased noise. PCA notably destabilized the optimal k, while the use of inverse weighting and Manhattan distance consistently led to more robust and higher-performing configurations.

When the conclusions are narrowed to what the confusion matrices themselves disclose (13Table 13.1 to 14Table 14.12), a clear and consistent pattern emerges across experiments: for NRW the *Unsatisfactory* class is almost uniformly the most reliably identified category, typically showing many true positives, while the *Medium* class is the recurrent source of error—often being misclassified as either *Good* or *Unsatisfactory*, which reveals a local overlap of classes in feature space that no single preprocessing choice entirely resolves. For RL the confusion matrices display a similar asymmetry but with nuances: *Good* cases are frequently the best recovered, *Medium* tends again to be the class most often misplaced (frequently labelled as *Good*), and *Unsatisfactory* shows variable performance, sometimes well detected and sometimes substantially confused with *Medium*. PCA runs stand out in the matrices by producing much more dispersed misclassifications, while experiments that removed redundant predictors tended to concentrate errors more narrowly and improved the clarity of extreme classes in many cases.

4 Naive Bayes

4.1 Data scope and candidate inputs

In this section we focus on 2023 data and restrict the sample to retail service utilities (`Tipo_Sistema` containing "baixa"), in line with the project statement that classification should be performed for the distribution-level providers (end-consumer supply).

Two multiclass targets are modeled independently:

- `NRW_class`: service quality classes for Non-Revenue Water (Good / Medium / Unsatisfactory),
- `RL_class`: service quality classes for Real Losses.

These classes correspond to ERSAR regulatory bands for the indicators AA08b (Non-Revenue Water, expressed as a percentage of water input to the system) and AA15b (Real Losses, expressed either as $\ell/(\text{connection} \cdot \text{day})$ or $\text{m}^3/(\text{km} \cdot \text{day})$, depending on service connection density). The goal is to reproduce, and ideally anticipate, regulatory classification.

As potential inputs we include:

- **All numeric columns** remaining after removing pure identifiers (`Empresa`, `Tipo_Sistema`) and the targets (`NRW_class`, `RL_class`);
- **Two categorical descriptors** (kept as factors): `V_Modelo_de_gestão` (management model) and `V_Tipologia_da_área_de_intervenção` (type of service area / territory).

This reflects an important modeling choice. We do not feed identifiers to the classifier because they allow trivial "memorization" of the utility, especially in distance-based models like KNN, and they have no operational meaning. Instead we focus on physical, operational and management descriptors.

Following the project brief, each task is studied under two settings: (i) *including* the task-specific key variables; (ii) *excluding* those key variables and selecting only from the remaining pool. This mirrors the requirement to evaluate models with and without the formula-defining inputs.

Task keys. These are precisely the variables that ERSAR itself uses to compute the indicator. They are, by design, the "obvious" predictors:

NRW: `V_Água_entrada_no_sistema` (water input to the system), `I_Água_não_faturada` (non-revenue water volume).

RL: `I_Perdas_reais_de_água` (real losses), `V_Comprimento_total_de_condutas` (total pipe length), `V_Ramais_de_ligação` (service connections).

The "with keys" regime asks: *Given the core physical quantities the regulator uses, can Naive Bayes recover the class?*

The "without keys" regime asks a stricter and more interesting question: *If those core quantities are missing or unreliable, can we still flag problematic utilities using indirect operational or managerial signals?*

4.2 Preprocessing pipeline (fold-wise, leakage-safe)

All transformations are fitted *inside* each CV training fold and applied to its validation fold only (i.e. nothing from the validation split is used to compute preprocessing statistics). This prevents information leakage and yields an honest estimate of generalization.

1. **Numerics:** per-column median imputation, then standardization using fold-wise mean and standard deviation (degenerate SDs replaced by 1 to avoid division by zero).
   Median imputation is robust to skew and outliers, which is relevant here because several infrastructure and cost indicators are heavy-tailed.
2. **Categoricals:** fixed level enumerations for `V_Modelo_de_gestão` and `V_Tipologia_da_área_de_intervenção`; missing/empty mapped to `__MISSING__`; unseen levels at validation mapped to `__NOVEL__`.
   This guarantees that (i) the model never "breaks" on an unseen category, and (ii) all folds use a consistent representation.

This fold-wise preprocessing is analogous in spirit to what we used for KNN in the previous chapter (standardization, imputation, handling of novel categorical levels), but is adapted here to a generative probabilistic model instead of a distance-based one.

4.3 Mixed-type Naive Bayes

We use `e1071::naiveBayes`. Numeric columns are modeled by class-conditional Gaussians (Gaussian Naive Bayes). Factor columns are modeled by class-conditional multinomials with Laplace smoothing (`laplace = 1`). This mixed formulation has two advantages:

- It natively supports heterogeneous data (continuous operational metrics + categorical descriptors) without one-hot expansion of factors, so it stays compact and less noisy.
- It yields full posterior class probabilities, which we will later evaluate with Multiclass LogLoss.

Note that Naive Bayes assumes conditional independence of features given the class and (for numerics) approximately normal within-class distributions. Neither assumption is perfect for water network data: pipe length, number of service connections and consumption volumes are strongly correlated; financial and infrastructure variables are often skewed. Even so, Naive Bayes is known to perform surprisingly well in practice because its parameter estimates are simple and low-variance, which matters with moderate sample sizes.

4.4 Cross-validation and quality control

We perform stratified 5-fold cross-validation and aggregate *out-of-fold* (OOF) predictions, so every observation is predicted by a model that did *not* see it during training. Quality is measured by three complementary metrics:

1. **Accuracy (Acc)**: overall correct rate. This can be optimistic if one class dominates.
2. **Balanced Accuracy (BA)**: mean of per-class recalls. This metric gives equal weight to Good, Medium and Unsatisfactory, which is essential here because the classes are not evenly represented. BA is therefore our primary selection criterion.
3. **Multiclass LogLoss**: negative log-likelihood of OOF probabilistic predictions (with small-ε clipping and row-wise renormalization to avoid numerical degeneracy). Low LogLoss means "confident and correct". High LogLoss means the model was confidently wrong.

When comparing candidate models we prioritize: higher BA $\rightarrow$ lower LogLoss $\rightarrow$ higher Acc $\rightarrow$ fewer features. This forces us to prefer models that (i) protect minority classes, (ii) are not overconfidently wrong, and (iii) remain simple.

4.5 Candidate screening by Mutual Information

Before fitting models we score all non-key predictors by their mutual information (MI) with the target class. Numeric variables are discretized by equal-frequency binning; categorical variables use their factor levels directly.

Intuition: MI quantifies how much knowing a variable reduces uncertainty about the class. High MI means "this variable is informative for distinguishing Good vs Medium vs Unsatisfactory". Appendix 7.3.1 shows the top-ranked variables (excluding the task's own keys) for each target. We observe:

- For **NRW_class**, governance and infrastructure knowledge matter: `V_Modelo_de_gestão`, `V_índice_de_conhecimento_infraestrutural` and `V_índice_de_gestão_patrimonial_de_infraestruturas` all score highly, suggesting that more structured, better-documented networks tend to have lower NRW. Financial variables such as `V_Rendimentos_tarifários` and `V_Gastos_totais` also appear, hinting that economic scale and billing structure co-move with NRW outcomes.
- For **RL_class**, physical loss volumes and resilience dominate: `V_Perdas_reais`, `I_Água_não_faturada`, `V_Água_não_faturada`, as well as `V_índice_de_segurança_e_resiliência` and failure-related indicators such as `I_Ocorrência_de_falhas_no_abastecimento`. This is consistent with the physical interpretation of Real Losses: structural leakage and service failures translate directly into worse RL class.

4.6 Variable selection experiments

We now build actual Naive Bayes models and measure out-of-fold (5-fold CV) performance while varying how many *extra non-key variables* (ranked by MI) we add on top of the task keys. We repeat this for both targets (`NRW_class`, `RL_class`) and for both regimes: with keys and without keys.

Balanced Accuracy (BA) is our primary quality criterion, followed by LogLoss and Accuracy (Acc).

4.6.1 With keys

When the key variables are included, the model is effectively allowed to "see" the same physical quantities used by ERSAR to construct the indicator. As expected, performance is very strong.

Top-3 Configurations — With Keys

(a) NRW_class (with key variables)

Extra variables	Acc	BA	LogLoss
2	0.923	**0.879**	0.2836
1	0.923	0.877	0.3309
3	0.908	0.855	0.3271

(b) RL_class (with key variables)

Extra variables	Acc	BA	LogLoss
0	0.908	**0.899**	1.1455
2	0.893	0.884	1.0633
1	0.872	0.856	1.0896

Confusion Matrices — With Keys

(a) NRW_class (best with keys: 2 extra variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	31	5	0
Medium	2	25	4
Unsatisfactory	0	4	125

(b) RL_class (best with keys: 0 extra variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	80	3	0
Medium	5	46	2
Unsatisfactory	3	5	52

These results show very high BA (≈ 0.88–0.90) and excellent class separation, especially for the *Unsatisfactory* class, which is the most operationally relevant from a regulatory standpoint (it flags poor performance).

4.6.2 Without keys

Here we explicitly *remove* the variables that are used to compute the ERSAR indicator. This scenario is more realistic for early-warning or partial-data monitoring: can we still guess a utility's class from indirect signals if some core volumes or lengths are missing?

Top-3 Configurations — Without Keys

(a) NRW_class (without key variables)

Selected variables	Acc	BA	LogLoss
6	0.776	**0.663**	0.9772
3	0.781	0.646	0.7211
7	0.724	0.638	1.4650

(b) RL_class (without key variables)

Selected variables	Acc	BA	LogLoss
2	0.541	**0.477**	0.9841
7	0.526	**0.477**	1.7866
4	0.536	0.473	1.2140

Confusion Matrices — Without Keys

(a) NRW_class (best without keys: 6 selected variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	27	5	4
Medium	5	11	15
Unsatisfactory	2	13	114

(b) RL_class (best without keys: 2 selected variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	76	4	3
Medium	29	7	17
Unsatisfactory	19	18	23

We see two important phenomena:

- The *Unsatisfactory* class is still often identified correctly (e.g. 114 out of 129 in NRW), which is encouraging from a regulatory monitoring perspective: even without direct access to the key variables, the worst performers tend to look structurally different.
- The *Medium* class is much harder. Misclassifications tend to send "Medium" utilities either down to "Unsatisfactory" or up to "Good", indicating that intermediate performance is less structurally stable and overlaps with both extremes.

Balanced Accuracy drops (e.g. from ≈ 0.88 to ≈ 0.66 for NRW, and from ≈ 0.90 to ≈ 0.48–0.52 for RL). This confirms that the ERSAR-defining variables are genuinely the main discriminators, but also shows that secondary variables still carry non-trivial signal.

4.7 Optimal Parameter Selection

Building on the variable-count ranges established above, we *loop* over a compact grid of Naive Bayes hyperparameters for each target and regime (with / without keys). For each selected variable count we evaluate all combinations of: **Laplace** $\in \{0, 1, 2\}$ (smoothing on categorical likelihoods, improving probability calibration), **threshold** $\in \{10^{-6}, 10^{-4}, 10^{-3}\}$ (floors tiny posteriors before renormalization to temper overconfident errors), and **eps** $\in \{0, 10^{-9}, 10^{-6}\}$ (floors Gaussian densities to prevent numerical explosions in LogLoss). We keep the same 5-fold stratified OOF protocol and the same ranking rule (maximize BA, then minimize LogLoss, then maximize Acc). The variable counts tested here are those found competitive in the previous section.

Final tuned models (best configuration per case).

- **NRW_class — with keys.** The best model uses the two key variables plus two extras (`V_Modelo_de_gestão`, `I_Perdas_reais_de_água`) with *Laplace*$= 1$, *threshold*$= 10^{-4}$, *eps*$= 10^{-9}$. It attains BA $= 0.881$, Acc $= 0.929$, LogLoss $= 0.190$. Compared to the untuned variant in the previous section, tuning mostly sharpens calibration (LogLoss drops from ≈ 0.28 to $\approx \mathbf{0.19}$) while keeping BA at the same high level.
- **RL_class — with keys.** The three key variables alone are sufficient. A representative optimum uses *Laplace*$= 2$, *threshold*$= 10^{-3}$, *eps*$= 10^{-6}$, yielding BA $= 0.915$, Acc $= 0.924$, LogLoss $= 0.508$. (Smoothing values 0–2 perform similarly here; the improvement versus the untuned model is mainly a large LogLoss reduction from ≈ 1.15 to $\approx \mathbf{0.51}$.)
- **NRW_class — without keys.** The best outcome is reached with six non-key variables (`V_Modelo_de_gestão`, `I_Perdas_reais_de_água`, `V_índice_de_conhecimento_infraestrutural`, `V_índice_de_gestão_patrimonial_de_infraestruturas`, `V_Tipologia_da_área_de_intervenção`, `V_Rendimentos_tarifários`) with *Laplace*$= 0$, *threshold*$= 10^{-3}$, *eps*$= 0$. Performance: BA $= 0.682$, Acc $= 0.781$, LogLoss $= 0.991$. Removing keys therefore creates a clear BA gap that tuning cannot close.
- **RL_class — without keys.** The best setting uses six non-key variables (`V_Perdas_reais`, `I_Água_não_faturada`, `V_Água_entrada_no_sistema`, `V_índice_de_segurança_e_resiliência`, `V_Modelo_de_gestão`, `I_Ocorrência_de_falhas_no_abastecimento`) with *Laplace*$= 1$, *threshold*$= 10^{-3}$, *eps*$= 0$. Performance: BA $= 0.516$, Acc $= 0.546$, LogLoss $= 1.561$.

Across both targets, the *with keys* regime consistently delivers substantially higher BA, confirming that the indicators encode essential signal; tuning mainly improves probability calibration (lower LogLoss) without altering the preferred variable counts. *The complete set of tuned configurations and the corresponding OOF confusion matrices are provided in the Appendix 7.3.2.*

4.8 Discussion

The Naive Bayes results answer two central questions posed in the project.

(1) Can we recover ERSAR's classification using the same core physical quantities? Yes. With the key variables included, Balanced Accuracy is high (often ≥ 0.88 for NRW and ≥ 0.90 for RL), and confusion matrices show that the *Unsatisfactory* class is especially well captured. This is not surprising: the keys *are* the mathematical ingredients of the indicators AA08b (NRW) and AA15b (RL). A probabilistic classifier fed those quantities will naturally recreate ERSAR's thresholds.

(2) Can we still say something useful without those key variables? Partially. When we exclude the keys, Balanced Accuracy drops (≈ 0.66 for NRW, ≈ 0.48–0.52 for RL), and the *Medium* class becomes hard to identify. Still, performance is *not* random: the model continues to correctly flag a large share of *Unsatisfactory* utilities, which is valuable for screening and early-warning. This means there is actionable signal in secondary variables such as infrastructure knowledge indices, management model, financial structure, failure/repair indicators and resilience metrics.

5 Comparative Analysis of KNN and Naive Bayes Models

The comparison between K-Nearest Neighbors (KNN) and Naive Bayes (NB) reveals complementary behaviors driven by their underlying assumptions and sensitivity to data characteristics. Both methods were evaluated under similar preprocessing regime, with and without inclusion of key variables, and using similar performance metrics.

Across both classification tasks, the Naive Bayes classifier consistently achieved higher Balanced Accuracy and substantially lower Log-Loss when key variables were included. For instance, the NB model reached BA ≈ 0.88 for NRW and BA ≈ 0.90 for RL, indicating strong probabilistic calibration and near-perfect recovery of ERSAR's regulatory classes. In contrast, the best KNN configuration for NRW achieved BA $= 0.842$, while the best for RL achieved BA $= 0.711$.

When key variables were excluded, both methods degraded in performance but in distinct ways. KNN exhibited a gradual decrease in accuracy, whereas NB suffered sharper declines in Balanced Accuracy (to ≈ 0.66 for NRW and ≈ 0.48–0.52 for RL). This difference stems from the fact that Naive Bayes relies on strong independence assumptions, so when key discriminative predictors are removed, the remaining correlated features violate its assumptions and reduce effectiveness. KNN, in contrast, still retains local distance information that can partially compensate for the loss of key predictors.

KNN's strength lies in its flexibility and ability to model non-linear class boundaries. However, its performance proved highly sensitive to data scaling, redundant predictors, and the choice of hyperparameters such as k, distance metric, and weighting scheme. The results showed that inverse weighting and Manhattan distance consistently stabilized performance and enhanced discrimination, particularly for NRW, which benefited from smaller k values when key variables were present. PCA, by contrast, consistently degraded results, likely because it disrupted the structure of distances between utilities.

Naive Bayes, being a generative and parametric model, produced much more stable results across folds and configurations. Its main advantage was interpretability and probabilistic calibration, yielding lower Log-Loss values even when the overall Balanced Accuracy decreased. Moreover, NB achieved near-perfect separation of extreme classes (Good vs. Unsatisfactory) when key variables were available, confirming that these inputs dominate ERSAR's regulatory thresholds. Nevertheless, its conditional independence assumption limited flexibility in capturing non-linear relationships, making intermediate classes (Medium) systematically harder to distinguish.

Overall, the comparative analysis shows that Naive Bayes provides stronger generalization and interpretability when key explanatory indicators are available, whereas KNN offers more flexible but less stable performance under feature reduction.

6 Conclusion

This study aimed to explore the determinants of non revenue water and real losses in Portuguese water utilities, using ERSAR's available performance data. Thus our work identified explanatory variables that capture inefficiencies in network performance and evaluated their predictive power through exploratory data analysis and classification modeling.

While the exploratory data analysis of real losses showed that rehabilitation variables seem to drive the largest changes between the years of 2020 and 2023, their overall explanatory power across utilities remains limited. On the other hand, the non revenue water indicator showed more promising results with a more clearer separation between performance classes, particularly for variables that express volumes relative to system input.

Additionally, the dataset used in this study was not extremely large, which imposes some restrictions on the generalization of the results. In the particular case of bulk type water utilities in Portugal, the small sample size restricts the ability to draw strong conclusions for this group. Future analyses could benefit from expanding the dataset by including more years or integrating additional sources of information.

From a methodological standpoint, both K-Nearest Neighbors and Naive Bayes yielded consistent yet complementary results. The Naive Bayes classifier achieved higher Balanced Accuracy and lower Log-Loss when key variables were included, reflecting stronger calibration and interpretability, whereas KNN delivered competitive accuracy and greater flexibility, particularly for the Non-Revenue Water classification. Nonetheless, KNN demonstrated higher sensitivity to preprocessing choices such as scaling, distance metrics, and correlated predictors, while Naive Bayes remained more stable but relied heavily on the presence of the core explanatory variables. Overall, Naive Bayes proved more effective when complete and well-structured data were available, whereas KNN offered a more adaptable alternative in noisier or partially observed scenarios.

Futhermore, future work could incorporate more discriminant features that better reflect operational efficiency. Operational data such as response time to pipe breaks and leakages, pressure levels within the distribution network and spatial concentration of failures could provide valuable insights, since water loss dynamics often depend on system pressure and on how quickly failures are addressed.

7 Appendix

7.1 Exploratory Data Analysis

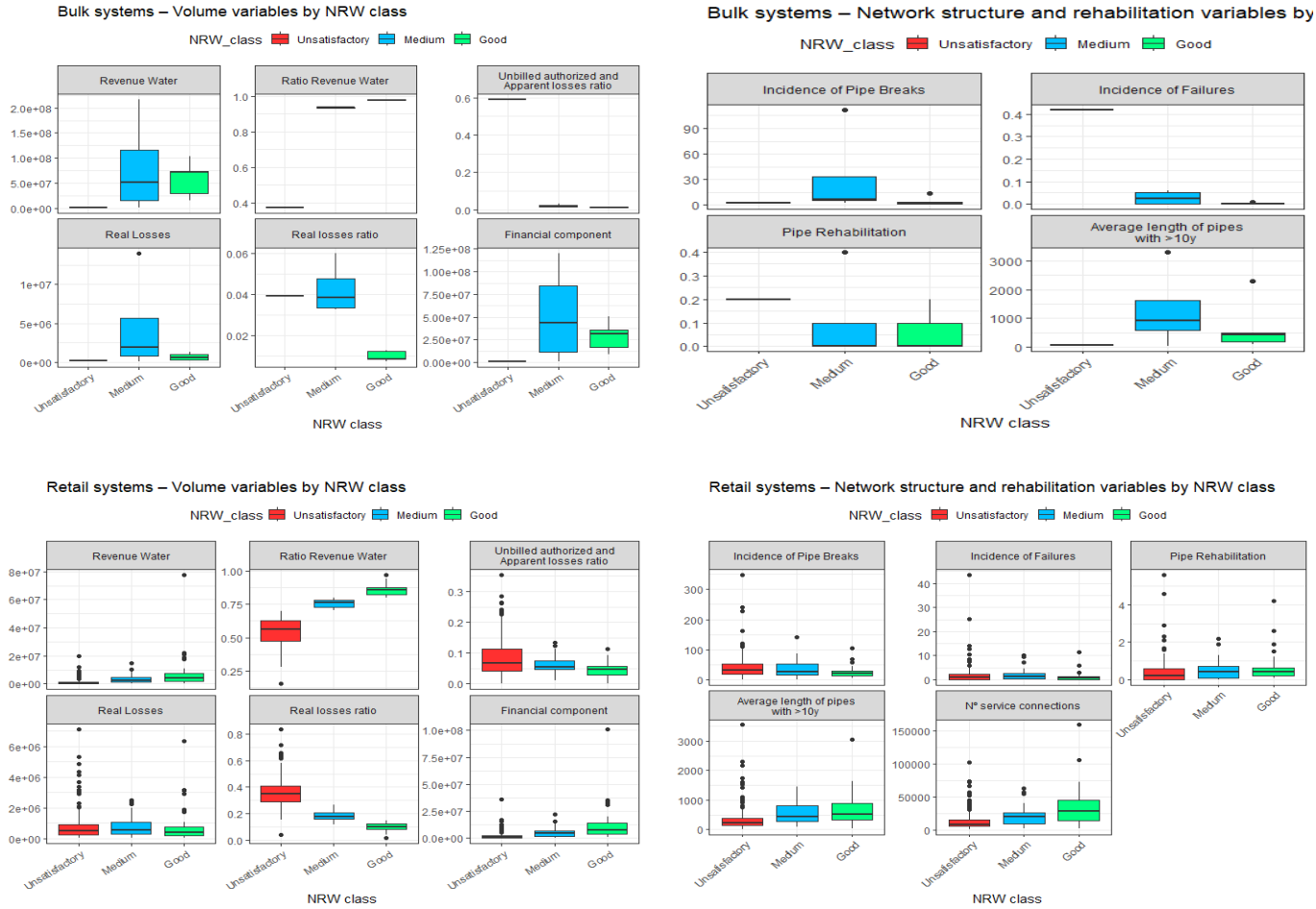


Figure 6: Boxplots of volume variables and variables related to network structure, rehabilitation and leakages across NRW classes

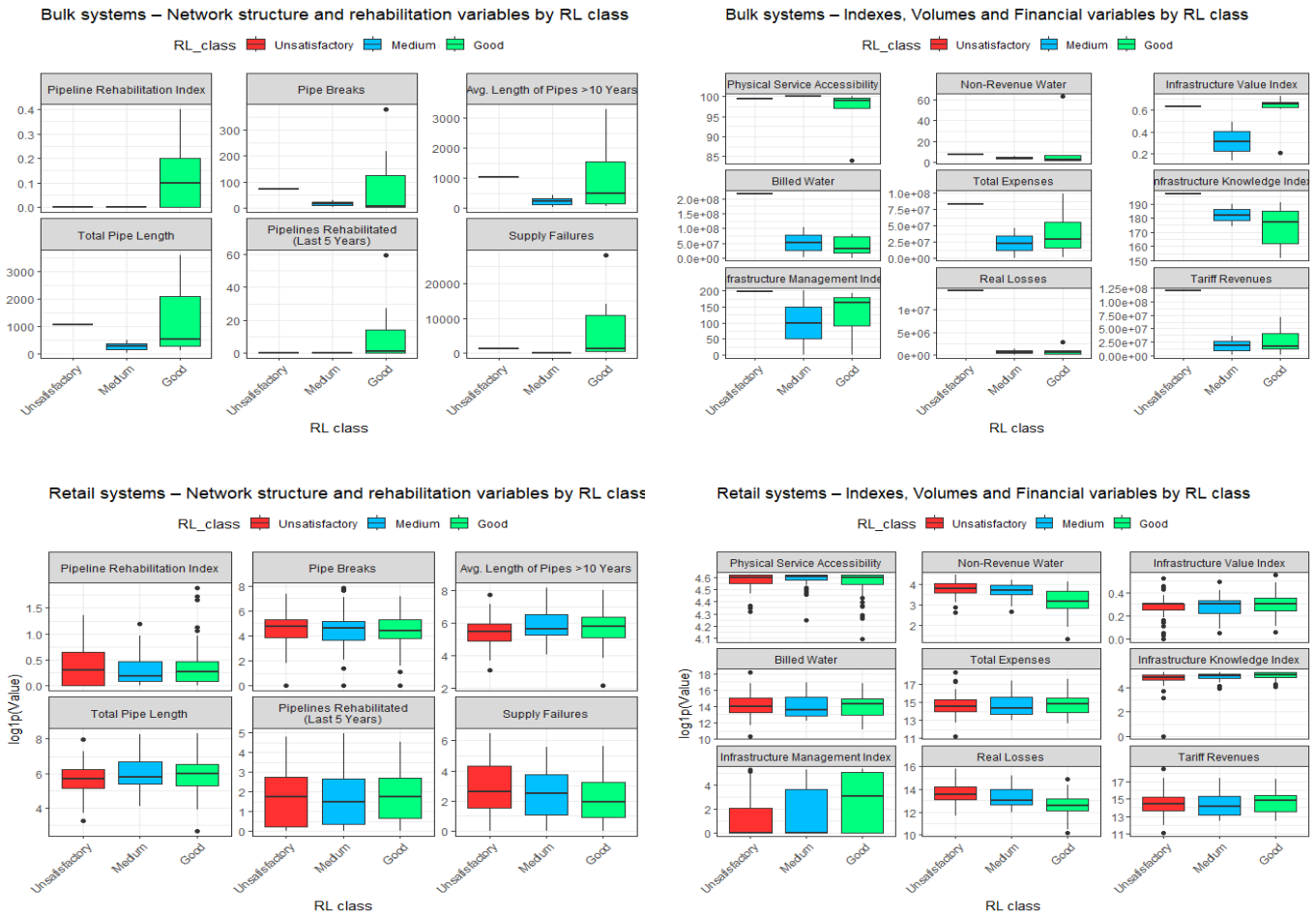


Figure 7: Boxplots of volume, indexes and financial variables and variables related to network structure, rehabilitation and leakages across RL classes

7.2 KNN Results Tables

7.2.1 Rectangular Weight

Table 7: Performance Summary – Rectangular Weight

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	31	0.659	0.419	0.716	0.592	0.734
NRW	Not Included	27	0.634	0.378	0.690	0.555	0.741
RL	Included	23	0.525	0.305	0.657	0.531	0.925
RL	Not Included	26	0.550	0.345	0.680	0.559	0.965

For NRW, including key variables improved overall performance ($\kappa = 0.419$, balanced accuracy $= 0.716$, F-measure $= 0.592$, log-loss $= 0.734$). For RL, however, removing them yielded slightly better classification metrics, suggesting redundancy or noise introduced by those variables.

7.2.2 Inverse Weight

Table 8: Performance Summary – Inverse Weight

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	42	0.707	0.472	0.733	0.629	0.721
NRW	Not Included	45	0.707	0.482	0.747	0.650	0.725
RL	Included	31	0.575	0.379	0.700	0.583	0.955
RL	Not Included	33	0.525	0.301	0.656	0.533	0.968

For NRW, performance was similar with and without key variables, though slightly higher balanced accuracy and F-measure were achieved without them. For RL, inclusion of key variables clearly improved all main metrics, confirming their predictive relevance.

7.2.3 Inverse Weight + PCA

Table 9: Performance Summary – Inverse Weight with PCA

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	15	0.415	0.0189	0.496	0.308	1.84
NRW	Not Included	34	0.634	0.323	0.654	0.512	0.939
RL	Included	38	0.250	-0.110	0.443	0.242	1.20
RL	Not Included	5	0.300	-0.039	0.476	0.290	7.21

Applying PCA before KNN drastically reduced performance in both tasks. For NRW, removing key variables improved metrics substantially, indicating that PCA may have discarded relevant variance. For RL, all configurations yielded poor results, with negative κ values implying near-random performance.

7.2.4 Inverse Weight + Removal of Highly Correlated Variables

Table 10: Performance Summary – Inverse Weight with Correlation Removal

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	15	0.732	0.526	0.779	0.686	2.24
NRW	Not Included	15	0.732	0.526	0.779	0.686	2.24
RL	Included	27	0.550	0.353	0.691	0.553	0.917
RL	Not Included	31	0.575	0.389	0.711	0.582	0.936

For NRW, results were identical with or without key variables, implying no added discriminative signal. For RL, slight improvements occurred after removing them, while the log-loss marginally increased, suggesting improved discrete classification but poorer probability calibration.

7.2.5 Inverse Weight + PCA + Correlation Removal

Table 11: Performance Summary – Inverse Weight with PCA and Correlation Removal

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	8	0.585	0.247	0.616	0.464	2.68
NRW	Not Included	8	0.585	0.247	0.616	0.464	2.68
RL	Included	36	0.325	0.009	0.506	0.321	1.27
RL	Not Included	7	0.300	-0.029	0.488	0.300	6.52

The combination of PCA and correlation filtering showed no improvement for NRW. For RL, including key variables slightly enhanced classification quality ($\kappa = 0.009$ vs. -0.029), but the overall performance remained weak with high log-loss values.

7.2.6 Inverse Weight + Manhattan Distance

Table 12: Performance Summary – Inverse Weight with Manhattan Distance

Target	Key Vars	Best k	Accuracy	κ	Bal. Accuracy	F-measure	Log-Loss
NRW	Included	15	0.780	0.625	0.842	0.739	0.590
NRW	Not Included	41	0.732	0.514	0.755	0.634	0.694
RL	Included	17	0.550	0.331	0.668	0.553	0.893
RL	Not Included	19	0.525	0.301	0.653	0.527	0.904

For NRW, inclusion of key variables clearly improved performance, particularly in κ (0.625), balanced accuracy (0.842), and F-measure (0.739). For RL, the improvement was modest yet consistent, indicating that these variables contribute useful information for both indicators.

7.2.7 Confusion Matrices – NRW

(Table 13.1) Rectangular Weight (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	2	0
Medium	3	3	7
Unsatisfactory	0	2	19

(Table 13.2) Rectangular Weight (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	4	2	0
Medium	4	3	7
Unsatisfactory	0	2	19

(Table 13.3) Inverse Weight (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	1	0
Medium	3	3	5
Unsatisfactory	0	3	21

(Table 13.4) Inverse Weight (without key vars)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	6	1	0
Medium	2	3	6
Unsatisfactory	0	3	20

(Table 13.5) Inverse Weight + PCA (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	3	5
Medium	3	0	7
Unsatisfactory	2	4	14

(Table 13.6) Inverse + PCA (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	4	2	4
Medium	2	2	2
Unsatisfactory	2	3	20

(Table 13.7) Inverse Weight + Removal of Highly Correlated Variables (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	0	1
Medium	2	5	5
Unsatisfactory	1	2	20

(Table 13.8) Inverse Weight + Removal of Highly Correlated Variables (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	0	1
Medium	2	5	5
Unsatisfactory	1	2	20

(Table 13.9) Inverse Weight + PCA + Correlation Removal (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	4	5
Medium	1	2	2
Unsatisfactory	4	1	19

(Table 13.10) Inverse Weight + PCA + Correlation Removal (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	4	5
Medium	1	2	2
Unsatisfactory	4	1	19

(Table 13.11) Inverse Weight + Manhattan Distance (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	1	2
Medium	1	5	4
Unsatisfactory	0	1	20

(Table 13.12) Inverse Weight + Manhattan Distance (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	2	1
Medium	1	2	4
Unsatisfactory	0	3	21

7.2.8 Confusion Matrices – RL

(Table 14.1) Rectangular Weight (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	1	1
Medium	9	6	3
Unsatisfactory	1	4	8

(Table 14.2) Rectangular Weight (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	1	0
Medium	9	7	4
Unsatisfactory	1	3	8

(Table 14.3) Inverse Weight (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	1	1
Medium	9	8	3
Unsatisfactory	1	2	8

(Table 14.4) Inverse Weight (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	7	2	1
Medium	9	6	3
Unsatisfactory	1	3	8

(Table 14.5) Inverse Weight + PCA (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	3	4
Medium	6	2	3
Unsatisfactory	8	6	5

(Table 14.6) Inverse Weight + PCA (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	4	2
Medium	8	1	4
Unsatisfactory	4	6	6

(Table 14.7) Inverse Weight + Removal of Highly Correlated Variables (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	1	0
Medium	10	9	4
Unsatisfactory	2	1	8

(Table 14.8) Inverse Weight + Removal of Highly Correlated Variables (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	5	1	0
Medium	11	9	3
Unsatisfactory	1	1	9

(Table 14.9) Inverse Weight + PCA + Correlation Removal (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	4	2
Medium	9	3	3
Unsatisfactory	5	4	7

(Table 14.10) Inverse Weight + PCA + Correlation Removal (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	3	3	4
Medium	9	5	4
Unsatisfactory	5	3	4

(Table 14.11) Inverse Weight + Manhattan Distance (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	9	2	1
Medium	7	6	4
Unsatisfactory	1	3	7

(Table 14.12) Inverse Weight + Manhattan Distance (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	8	2	0
Medium	6	6	5
Unsatisfactory	3	3	7

7.3 Naive Bayes

7.3.1 Mutual Information

NRW_class — Top-15 MI among non-keys

Feature	MI
V_Modelo_de_gestão	0.2677
I_Perdas_reais_de_água	0.1963
V_índice_de_conhecimento_infraestrutural	0.1949
V_índice_de_gestão_patrimonial_de_infraestruturas	0.1824
V_Tipologia_da_área_de_intervenção	0.1814
V_Rendimentos_tarifários	0.1790
V_Outros_rendimentos	0.1733
V_Água_faturada	0.1681
V_índice_de_medição_de_caudais	0.1652
I_Cobertura_dos_gastos	0.1653
V_Consumo_faturado_medido	0.1629
V_Água_faturada_não_doméstica	0.1605
V_Gastos_totais	0.1574
V_Água_faturada_doméstica	0.1570
V_Consumo_autorizado	0.1416

RL_class — Top-15 MI among non-keys

Feature	MI
V_Perdas_reais	0.1643
I_Água_não_faturada	0.1471
V_Água_não_faturada	0.1116
V_Modelo_de_gestão	0.1071
V_Água_entrada_no_sistema	0.0996
V_índice_de_segurança_e_resiliência	0.0848
I_Ocorrência_de_falhas_no_abastecimento	0.0846
V_índice_de_conhecimento_infraestrutural	0.0619
V_índice_de_medição_de_caudais	0.0590
V_Pessoal_afeto_ao_serviço_de_abastecimento_de_água	0.0557
V_índice_de_gestão_patrimonial_de_infraestruturas	0.0513
V_Água_faturada_doméstica	0.0494
V_Alojamentos_com_serviço_disponível_não_efetivo	0.0467
I_Acessibilidade_económica_do_serviço	0.0423
I_Ocorrência_de_avarias_em_condutas	0.0420

7.3.2 Naive Bayes Results Tables

Top-3 Tuned Configurations — With Keys

NRW_class (with key variables)

Extra	Laplace	Threshold	Eps	Acc	BA	LogLoss
2	1	10^{-4}	10^{-9}	0.9286	**0.8814**	0.1897
2	1	10^{-3}	10^{-9}	0.9286	**0.8814**	0.1912
2	0	10^{-4}	10^{-9}	0.9286	**0.8814**	0.1981

RL_class (with key variables)

Extra	Laplace	Threshold	Eps	Acc	BA	LogLoss
0	2	10^{-3}	10^{-6}	0.9235	**0.9153**	0.5084
0	1	10^{-3}	10^{-6}	0.9235	**0.9153**	0.5084
0	0	10^{-3}	10^{-6}	0.9235	**0.9153**	0.5084

Confusion Matrices — With Keys

NRW_class (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	31	4	1
Medium	2	25	4
Unsatisfactory	0	3	126

RL_class (with key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	81	2	0
Medium	4	47	2
Unsatisfactory	2	5	53

Top-3 Tuned Configurations — Without Keys

NRW_class (without key variables)

Selected	Laplace	Threshold	Eps	Acc	BA	LogLoss
6	0	10^{-3}	0	0.7806	**0.6818**	0.9910
6	0	10^{-4}	0	0.7806	**0.6818**	0.9995
6	0	10^{-6}	0	0.7806	**0.6818**	1.0225

RL_class (without key variables)

Selected	Laplace	Threshold	Eps	Acc	BA	LogLoss
6	1	10^{-3}	0	0.5459	**0.5157**	1.561
6	1	10^{-4}	0	0.5459	**0.5157**	1.561
6	1	10^{-6}	0	0.5459	**0.5157**	1.561

Confusion Matrices — Without Keys

NRW_class (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	27	5	4
Medium	4	13	14
Unsatisfactory	3	13	113

RL_class (without key variables)

Prediction / Truth	Good	Medium	Unsatisfactory
Good	64	13	6
Medium	18	27	8
Unsatisfactory	12	32	16